

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q18-1070

Battlefield 138 kV
May 25, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1070. The Interconnection Customer proposes a 340.6 MW natural gas combined cycle facility interconnecting at the Battlefield 138 kV substation in the Virginia area. The request was received on September 29, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 339 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$5,886,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

TOANO / LIGHTFOOT 138 kV: 152.9% loading (335 MVA) following the outage of the MIRA VISTA / YARNEN SUBSTATION 345 kV line; DFAX = 53.2%.

SYNATA HUGOTON / LIGHTFOOT 138 kV: 145.9% loading (376 MVA) following the outage of the MIRA VISTA / YARNEN 345 kV; DFAX = 53.2%.

For the loss of MIRA VISTA / YARNEN 345 kV line, the SYNATA HUGOTON / BATTLEFIELD 138 kV loads to 141.3% of its 409 MVA rating. Project DFAX: 53.2%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
TOANO / LIGHTFOOT 138 kV	Rebuild line with higher-capacity conductor	\$2,540,000	49% (\$1,249,000)
SYNATA HUGOTON / LIGHTFOOT 138 kV	Rebuild line with higher-capacity conductor	\$7,136,000	49% (\$3,510,000)
SYNATA HUGOTON / BATTLEFIELD 138 kV	Rebuild line with higher-capacity conductor	\$2,292,000	49% (\$1,127,000)
	Total allocated to project		\$5,886,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.